

Jeel Tandel

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Summary

Backend Software Engineer and recent graduate with production experience in **Python (FastAPI)** and **Java**. Engineered a RAG-based AI search engine and optimized **PostgreSQL** queries to improve reporting speeds by 20%. Passionate about database internals, API architecture, and high-performance backend systems.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL (PL/SQL)

Backend & APIs: FastAPI, Node.js, RESTful Architecture, OAuth 2.0, Microservices

Data & Database: PostgreSQL, Oracle 19c, MongoDB, Pinecone (Vector DB), Redis, Pandas

DevOps & Tools: Docker, Git/GitHub, Oracle Cloud (OCI), Linux, Jira, LangChain

Professional Experience

AI Software Engineer, Xenara AI

March 2025 – January 2026

- **Engineered** a production-ready Retrieval-Augmented Generation (RAG) pipeline using **LangChain** and **Pinecone**, enabling semantic search across large unstructured datasets.
- **Built** high-concurrency **FastAPI** endpoints integrated with **Redis** caching, reducing data retrieval latency by **30%** for high-traffic routes.
- **Deployed** containerized applications using **Docker** on Oracle Cloud Infrastructure (OCI), automating the transition from development to production environments.
- **Automated** ETL workflows using Python and SpaCy to extract named entities from raw text, transforming unstructured inputs into structured JSON for downstream analytics.

Full-Stack Developer Intern, Appy.yo

September 2024 – December 2024

- **Diagnosed** and resolved SQL query bottlenecks by restructuring **PostgreSQL** schemas and indexes, resulting in a **20% execution speed improvement** for critical internal reporting tools.
- **Implemented** secure **REST API** components using **Java** and Python, integrating OAuth 2.0 authentication flows to safeguard multi-tenant user data.
- **Shipped** code in 2-week Agile sprints, conducting peer code reviews to enforce strict typing and design patterns.

Software Projects

NASA Space Apps: Flood Risk Analyzer / Python, GIS APIs, Pandas

Fall 2024

- **Built** a real-time flood visualization prototype in a 48-hour hackathon sprint, integrating multiple environmental APIs to map critical risk data.
- **Engineered** data cleaning pipelines using **Pandas/GeoPandas** to validate spatial data accuracy before rendering, handling edge cases in API responses.

Enterprise Music Database (Spotify Clone) / Oracle 19c, PL/SQL

Winter 2025

- **Designed** a normalized (3NF) relational database schema to handle complex Many-to-Many relationships between artists, albums, and users.
- **Wrote** PL/SQL stored procedures and triggers to automate backend logic for playlist generation, ensuring data consistency without application-layer overhead.

Distributed Task Manager / Java, TypeScript, REST API

Spring 2025

- **Developed** a distributed REST API layer using **Java**, achieving **90%+ unit test coverage** to ensure stability during scaling.
- **Constructed** a responsive TypeScript frontend that consumes backend services for real-time task tracking and status updates.

Education

Advanced Diploma, Computer Programming & Analysis, Algonquin College

Graduating: Aug 2025

GPA: 3.7/4.0 | **Dean's Honors List (All Terms)**

Relevant Coursework: Database Design (Oracle/SQL), Data Structures & Algorithms, Systems Analysis.

Certifications & Awards

Certifications: Oracle Cloud Infrastructure (OCI) AI Foundations Associate (2025), OCI Foundations Associate (2025)

Hackathon: NASA Space Apps Challenge Participant (Fall 2024)

Leadership: Class Representative (Liaison for 50+ students)